Jiajun Hu

(412) 726-8574 | jiajunh5@asu.edu | codingmmax.github.io

Education

Arizona State University — Tempe, AZ
Ph.D. in Computer Engineering — 08/2024 – Present

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering — 02/2022 – 12/2024

University of Nottingham, Ningbo, China — Ningbo, China
Bachelor of Engineering in Electrical and Electronics Engineering — 09/2017 – 07/2021

Experience

Research Associate Intern — HPE Labs — Fort Collins, CO
Emerging Accelerator Team — 05/2026 – Present

- Developing FPGA-ReRAM chiplet architectures for efficient LLM inference and memory-intensive acceleration.
- Exploring the design space of ReRAM-IMC-enhanced FPGA chiplets for emerging LLM workloads.

Skills

Programming: C/C++, Python, SystemVerilog, CUDA
Tools: Verilog-to-Routing (VTR), Synopsys Design Compiler, Verilator, Git

Research Summary

FPGA Architectures with In-/Near-Memory Compute for Deep Learning

- ReRAM-enhanced FPGA architectures with design-space exploration and mapping for deep learning.
- Chiplet-based FPGA-ReRAM architectures for LLM inference.
- Direct BRAM-DSP paths with placement-aware CAD extensions.

Efficient Mapping and Analytical Frameworks for Compute-in-Memory Architectures

- Fast automated mapping for compute-in-DRAM architectures.
- Reuse-aware GEMM/GEMV performance modeling for compute-in-DRAM architectures.

Selected Projects

***NIFA*: Non-Linear IMC Enhanced FPGA for Efficient ML Inference**

- Designed an FPGA architecture with ReRAM-based in-memory computing (IMC) blocks providing in-block nonlinear function support.
- Accelerated attention computation via a log/exp-based decomposition, exploiting the nonlinear capability of the ReRAM IMC blocks.

***RACAM*: Reuse-Aware In-DRAM Bit-Serial PIM with Automated Mapping**

- Designed and implemented an in-DRAM bit-serial analytical performance simulator supporting end-to-end LLM workloads across a range of GEMM/GEMV kernels.
- Extended the simulator to enable flexible GEMM/GEMV mapping schemes optimized for DRAM architectures, determining the optimal data layout for each LLM kernel to enhance DRAM-PIM efficiency.
- Performed area estimation of DRAM-integrated peripheral units using Synopsys Design Compiler.

Publications

- **Jiajun Hu**, Ruthwik Reddy Sunketa, Lei Zhao, Archit Gajjar, Luca Buonanno, and Aman Arora. *NIFA: Nonlinear IMC Enhanced FPGA for Efficient ML Inference*. In *Proc. FPL 2026* (to appear). arXiv:2607.15123
- Siyuan Ma, **Jiajun Hu**, Jeeho Ryoo, Aman Arora, and Lizy Kurian John. *RACAM: Enhancing DRAM with Reuse-Aware Computation and Automated Mapping for ML Inference*. In *Proc. PACT 2026* (to appear). arXiv:2512.09304
- **Jiajun Hu**, Chetan Choppali Sudarshan, Maxwell Clifford, Vidya Chhabria, and Aman Arora. *CarbonSet: A Dataset to Analyze Trends and Benchmark the Sustainability of CPUs and GPUs*. In *Proc. GLSVLSI '25*, pp. 177–184. doi:10.1145/3716368.3735235
- Chetan Choppali Sudarshan, **Jiajun Hu**, Aman Arora, and Vidya A. Chhabria. *CarbonPATH: Carbon-aware pathfinding and architecture optimization for chiplet-based AI systems*. arXiv preprint (under review). arXiv:2603.03878
- **Jiajun Hu**, Ruthwik Reddy Sunketa, Andrew Boutros, and Aman Arora. *Boosting FPGA Performance with Direct BRAM-DSP Paths*. arXiv preprint (under review). arXiv:2607.05756